

Assignment 3

Name: Diya Jain U81673988

Q1. (5 points) Assume that the following section of main memory is used to store the page table for 3 different processes. The page-table base register values for process P1 is 1080, for P2 is 1085, and for P3 is 1090. Assume that the contents of memory below correspond to frame numbers. Also assume that frame size is 4096.

contents	3	5	8	4	7	2	0	15	11	18	6	20	24	9	10	13	30	38	40	1
Memory addresses	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	7	7	7	7	8	8	8	8	8	8	8	8	8	8	9	9	9	9	9	9
	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5

To which physical memory address would the logical address (1, 1200) correspond to if generated by P1?

$$(2 \times 4096) + 1200$$

Answer: **9392**

To which physical memory address would the logical address (3, 800) correspond to if generated by P2?

$$(23 \times 4096) + 800$$

Answer: **95008**

Assume that each process has 5 pages. To which process does the following physical address belong and which logical address corresponds to each physical address:

$$\text{Frame Number} = (57344 / 4096) = 14$$

$$\text{Offset} = 57344 - (4096 * 14) = 0$$

The address at which the frame number “14” resides is “1091” in process P3

The page table base register value for P3 is “1090”

$$\text{The Page Number} = \text{address} - \text{base register value} = 1091 - 1090 = 1$$

Physical address 57344 **Process P3 Logical address (1,0)**

$$\text{Frame Number} = (57343 / 4096) = 13$$

$$\text{Offset} = 57343 - (4096 * 13) = 4095$$

The address at which the frame number “13” resides is “1087” in process P2

The page table base register value for P2 is “1085”

$$\text{The Page Number} = \text{address} - \text{base register value} = 1087 - 1085 = 2$$

Physical address 57343 **Process P2 Logical address (2,4095)**

Q2. (5 points) A paging system is experiencing a page fault rate of 1 in 1 million page references. When a page fault occurs, 30% of the time an empty frame is not available and frame replacement is needed. Assume that it takes 10 milliseconds to service a page fault needing no frame replacement. When frame replacement is needed, 40% of time the frame has been modified and it takes 20 milliseconds to service the page fault, in contrast it takes 12 milliseconds when the frame to be replaced has not been modified. It takes 100 nanoseconds to reference a physical memory location and we can neglect the time that it takes to access the page table. Calculate the effective memory access time for this system under these conditions. Clearly indicate intermediate steps.

Answer:

$$\text{Time} = 1 \text{ page fault time} + (10^6 - 1) \text{ page references}$$

$$\begin{aligned} 1 \text{ page fault time} &= 0.3(0.4(20 * 10^{-3}) + 0.6(12 * 10^{-3})) + 0.7(10 * 10^{-3}) \\ &= 0.3((8 + 7.2) * 10^{-3}) + 0.7(10 * 10^{-3}) \\ &= 0.3(15.2 * 10^{-3}) + 0.7(10 * 10^{-3}) \\ &= (4.56 + 7) * 10^{-3} \\ &= 11.56 * 10^{-3} \text{ sec} \end{aligned}$$

$$(10^6 - 1) \text{ page references} = (10^6 - 1) * 100 * 10^{-9} = 10^6 * 10^2 * 10^{-9} = 10^{-1} \text{ sec}$$

$$\text{Total Time} = 0.01156 + 0.1 = 0.11156 \text{ sec}$$

$$\begin{aligned} \text{Time per one reference} &= 0.11156 * 10^{-6} \\ &= 111.56 * 10^{-9} \\ &= \mathbf{111.56 \text{ nanoseconds}} \end{aligned}$$

$$p = (1 / 1000000) = 1 * 10^{-6}$$

$$\begin{aligned} \text{EAT} &= 1 * 10^{-6} * 10^6 (0.7 * 10 + 0.3(0.4 * 20 + 0.6 * 12)) \text{ (nanoseconds)} + 100 \text{ (nanoseconds)} \\ &= (11.56) + 100 = \mathbf{111.56 \text{ (nanoseconds)}} \end{aligned}$$